

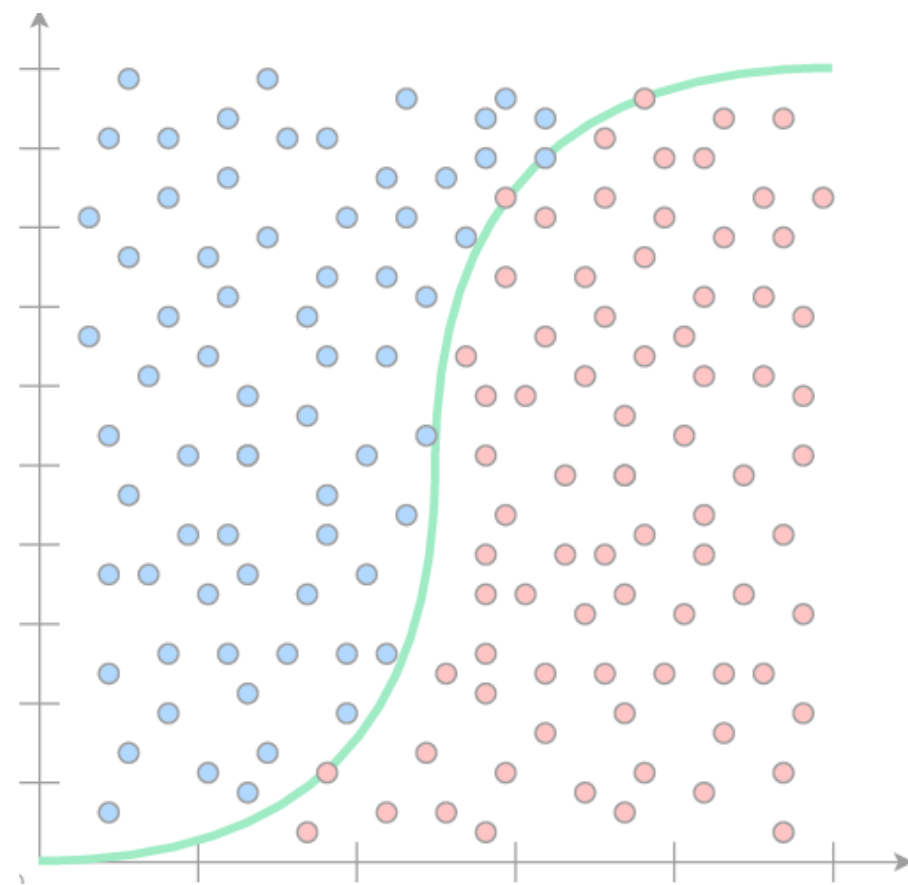
Lecture 3: Sequence Modeling



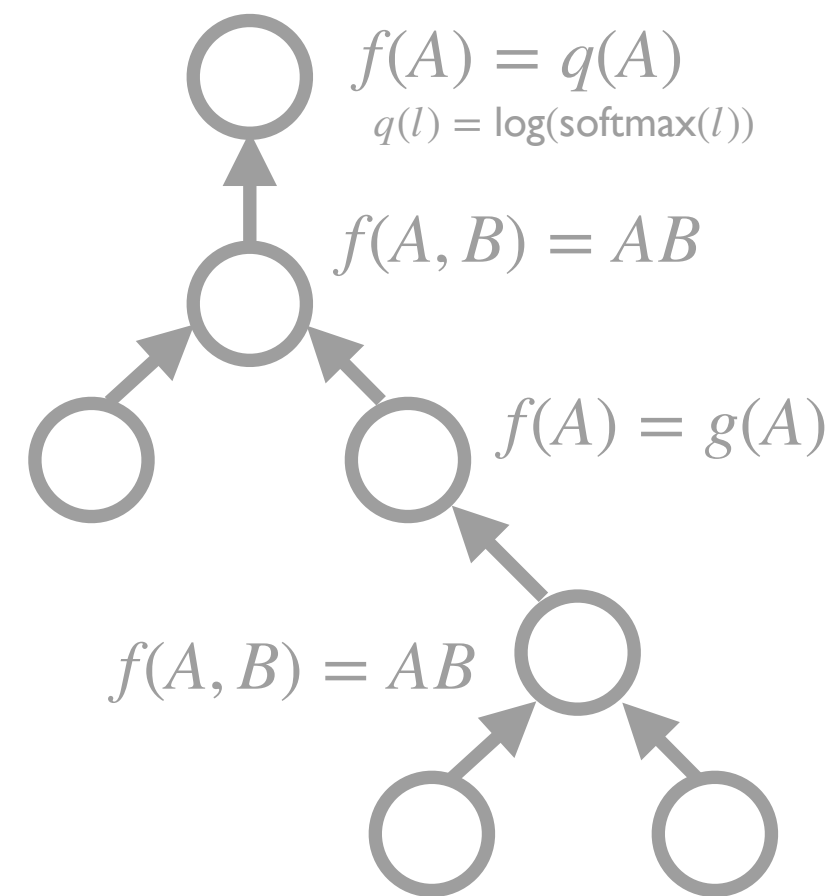
NYU

Based on slides by
Eunsol Choi

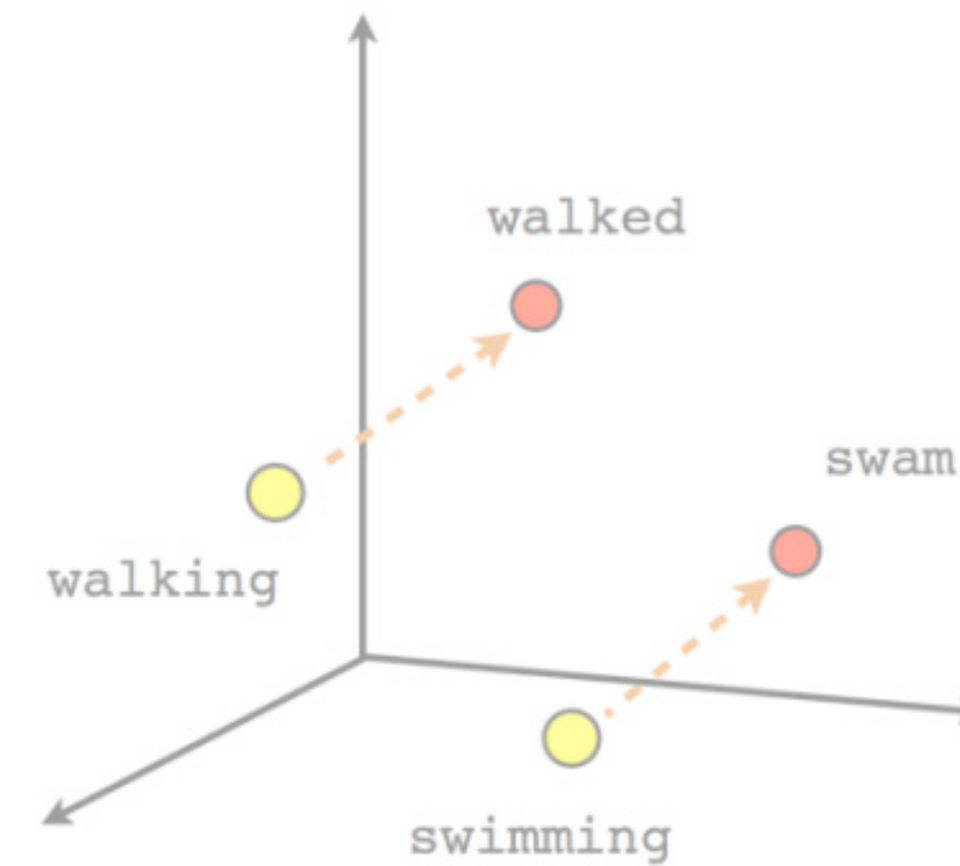
Overview



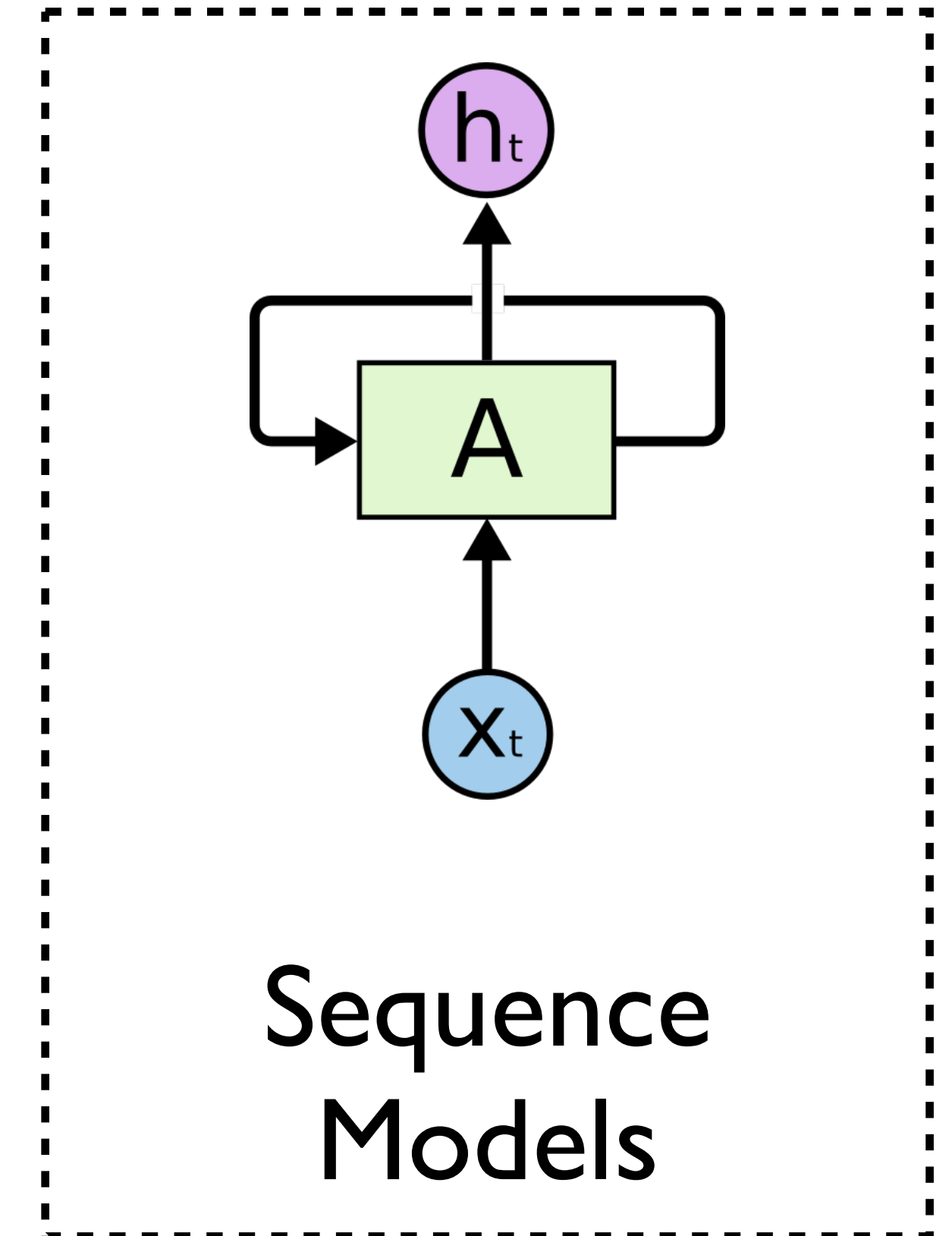
Logistic
Regression



Neural
Networks



Word
Embeddings



Sequence
Models

Today

- ▶ Quiz
- ▶ Sequence Modeling

Why Sequence Modeling?

- ▶ Consider the following examples for sentiment classification:
 - *How can you not see this movie?*
 - *You should not see this movie.*
- ▶ Would unigram, bigram models work?
- ▶ Would feedforward neural network (deep averaging network) work?
- ▶ We need a model to maintain a state to capture influences among words

Sequence Modeling

- ▶ **Sequence Modeling in NLP**
- ▶ Language Modeling Task
- ▶ Recurrent Neural Network
- ▶ Attention
- ▶ Transformer

The Structure of Language

I ate the spaghetti

I ate the spaghetti

The Structure of Language



I ate the spaghetti with chopsticks

The diagram illustrates the hierarchical structure of the sentence "I ate the spaghetti with chopsticks". It uses curved arrows to show the nesting of phrases. A long orange arrow starts at the beginning of the sentence and points to the word "chopsticks", indicating the main clause. A black arrow starts at "I" and points to "ate", indicating the subject and verb. Another black arrow starts at "the" and points to "spaghetti", indicating the object. A final black arrow starts at "with" and points to "chopsticks", indicating the modifier.



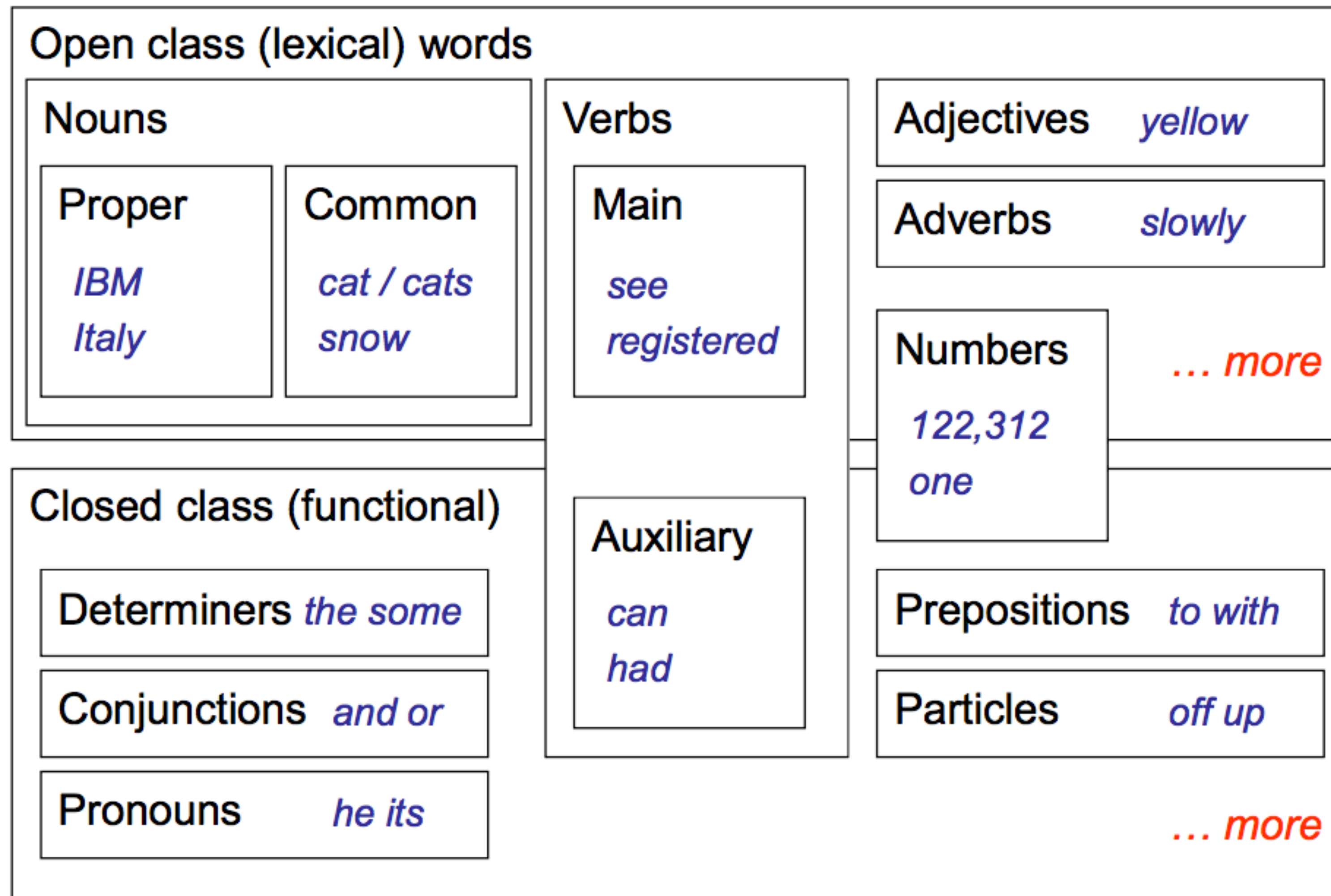
I ate the spaghetti with meatballs

The diagram illustrates the hierarchical structure of the sentence "I ate the spaghetti with meatballs". It uses curved arrows to show the nesting of phrases. A long orange arrow starts at the beginning of the sentence and points to the word "meatballs", indicating the main clause. A black arrow starts at "I" and points to "ate", indicating the subject and verb. Another black arrow starts at "the" and points to "spaghetti", indicating the object. A final black arrow starts at "with" and points to "meatballs", indicating the modifier.

- ▶ Language is tree-structured!
- ▶ Classifying words into grammatical categories (part-of-speech) can demonstrate that structure

Parts of Speech

- Categorization of words into types



Sequence Modeling Problems in NLP

- Parts of Speech Tagging (POS)

PRP VBZ DT NN IN NNS

I ate the spaghetti with chopsticks

PRP VBZ DT NN IN NNS

I ate the spaghetti with meatballs

CC	conjunction, coordinating	and both but either or
CD	numeral, cardinal	mid-1890 nine-thirty 0.5 one
DT	determiner	a all an every no that the
EX	existential there	there
FW	foreign word	gemeinschaft hund ich jeux
IN	preposition or conjunction, subordinating	among whether out on by if
JJ	adjective or numeral, ordinal	third ill-mannered regrettable
JJR	adjective, comparative	braver cheaper taller
JJS	adjective, superlative	bravest cheapest tallest
MD	modal auxiliary	can may might will would
NN	noun, common, singular or mass	cabbage thermostat investment subhumanity
NNP	noun, proper, singular	Motown Cougar Yvette Liverpool
NNPS	noun, proper, plural	Americans Materials States
NNS	noun, common, plural	undergraduates bric-a-brac averages
POS	genitive marker	' 's
PRP	pronoun, personal	hers himself it we them
PRP\$	pronoun, possessive	her his mine my our ours their thy your
RB	adverb	occasionally maddeningly adventurously
RBR	adverb, comparative	further gloomier heavier less-perfectly
RBS	adverb, superlative	best biggest nearest worst
RP	particle	aboard away back by on open through
TO	"to" as preposition or infinitive marker	to
UH	interjection	huh howdy uh whammo shucks heck
VB	verb, base form	ask bring fire see take
VBD	verb, past tense	pleaded swiped registered saw
VBG	verb, present participle or gerund	stirring focusing approaching erasing
VCN	verb, past participle	dilapidated imitated reunified unsettled
VBP	verb, present tense, not 3rd person singular	twist appear comprise mold postpone
VBZ	verb, present tense, 3rd person singular	bases reconstructs marks uses
WDT	WH-determiner	that what whatever which whichever
WP	WH-pronoun	that what whatever which who whom
WP\$	WH-pronoun, possessive	whose
WBB	Wh-adverb	however whenever where why

POS Tagging

- ▶ Many words have more than one POS, depending on its context

The back door = JJ (Adjective)

On my back = NN (Noun)

Win the voters back = RB (Adverb)

Promised to back the bill = VB (Verb)

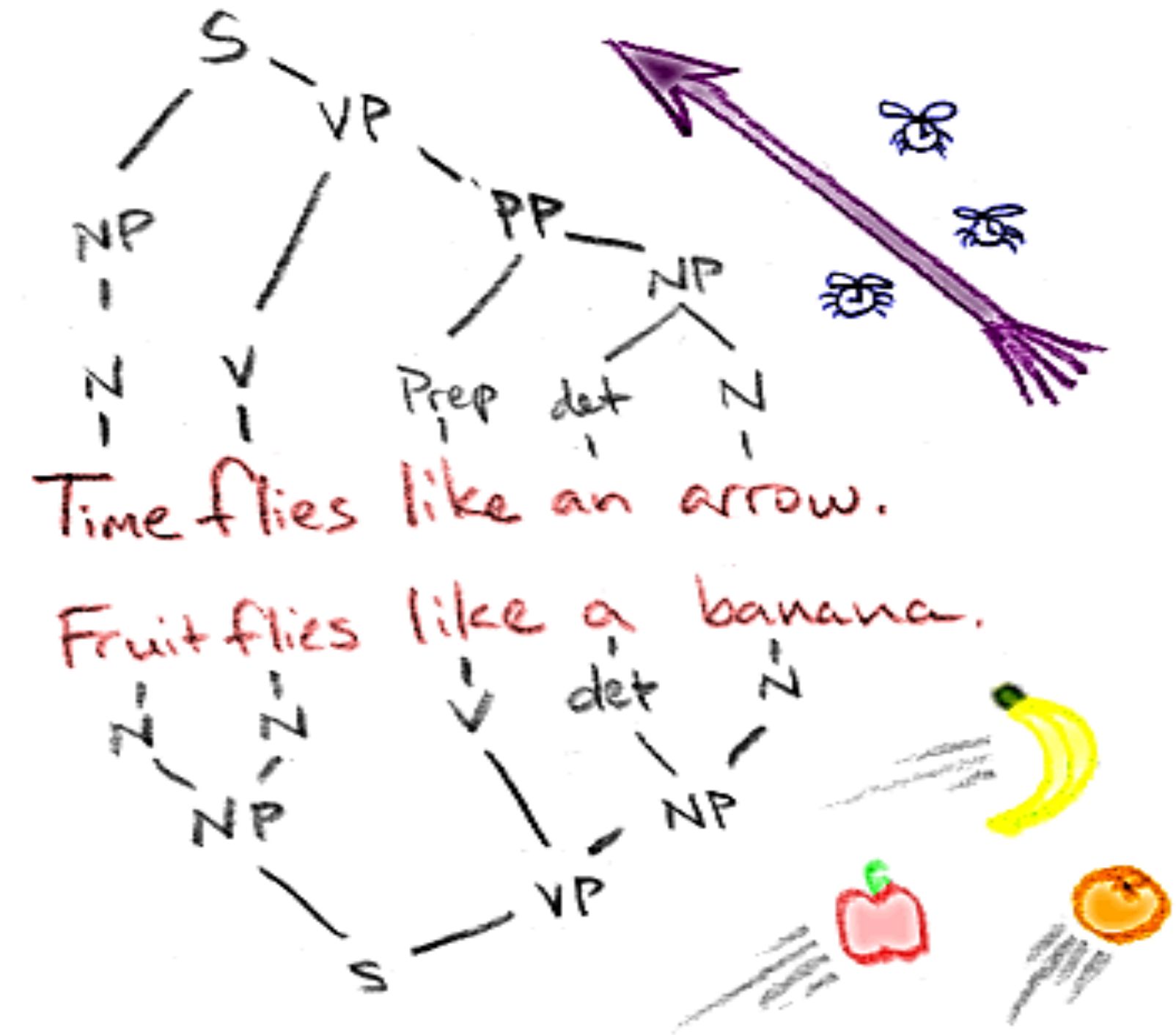
- ▶ The POS tagging problem is to determine the POS tag for a particular instance of a word.

Sources of Information

- Knowledge of neighboring words

Time flies like an arrow;
Fruit flies like a banana

- Knowledge of word probabilities
 - the, a, an*** is almost always article
 - man*** is frequently noun, rarely used as a verb
 - If we choose the most frequent tag, over 90% accuracy



What is this good for?

- ▶ Preprocessing step for syntactic parsers
- ▶ Domain-independent disambiguation for other tasks
- ▶ (Very) shallow information extraction:
 - ▶ write regular expressions like (Det) Adj*N + over the output for phrases

POS tag sets in different languages

Language	Source	# Tags
Arabic	PADT/CoNLL07 (Hajič et al., 2004)	21
Basque	Basque3LB/CoNLL07 (Aduriz et al., 2003)	64
Bulgarian	BTB/CoNLL06 (Simov et al., 2002)	54
Catalan	CESS-ECE/CoNLL07 (Martí et al., 2007)	54
Chinese	Penn Chinese Treebank 6.0 (Palmer et al., 2007)	34
Chinese	Sinica/CoNLL07 (Chen et al., 2003)	294
Czech	PDT/CoNLL07 (Böhmová et al., 2003)	63
Danish	DDT/CoNLL06 (Kromann et al., 2003)	25
Dutch	Alpino/CoNLL06 (Van der Beek et al., 2002)	12
English	Penn Treebank (Marcus et al., 1993)	45
French	French Treebank (Abeillé et al., 2003)	30
German	Tiger/CoNLL06 (Brants et al., 2002)	54
German	Negra (Skut et al., 1997)	54
Greek	GDT/CoNLL07 (Prokopidis et al., 2005)	38
Hungarian	Szeged/CoNLL07 (Csendes et al., 2005)	43
Italian	ISST/CoNLL07 (Montemagni et al., 2003)	28
Japanese	Verbmobil/CoNLL06 (Kawata and Bartels, 2000)	80
Japanese	Kyoto4.0 (Kurohashi and Nagao, 1997)	42
Korean	Sejong (http://www.sejong.or.kr)	187
Portuguese	Floresta Sintá(c)tica/CoNLL06 (Afonso et al., 2002)	22
Russian	SynTagRus-RNC (Boguslavsky et al., 2002)	11
Slovene	SDT/CoNLL06 (Džeroski et al., 2006)	29
Spanish	Ancora-Cast3LB/CoNLL06 (Civit and Martí, 2004)	47
Swedish	Talbanken05/CoNLL06 (Nivre et al., 2006)	41
Turkish	METU-Sabancı/CoNLL07 (Ofłazer et al., 2003)	31

[Petrov et al. 2012]

Universal POS Tag Set

Gillick et al. 2016

Language	CRF+	CRF
Bulgarian	97.97	97.00
Czech	98.38	98.00
Danish	95.93	95.06
German	93.08	91.99
Greek	97.72	97.21
English	95.11	94.51
Spanish	96.08	95.03
Farsi	96.59	96.25
Finnish	94.34	92.82
French	96.00	95.93
Indonesian	92.84	92.71
Italian	97.70	97.61
Swedish	96.81	96.15
AVERAGE	96.04	95.41

- Universal POS tagset (~12 tags), cross-lingual model works well!

Sequence Modeling Problems in NLP

- ▶ Named Entity Recognition (NER):
 - ▶ Segment text into spans with certain properties (person, organization, ..)

[Germany]_{LOC} 's representative to the [European Union]_{ORG} 's veterinary committee [Werner Zwingman]_{PER} said on Wednesday consumers should...

- ▶ Tagging each word tagging with BIO scheme
 - ▶ B: token that begins a span
 - ▶ I: token inside the span
 - ▶ O: tokens outside of the span

Germany/**BL** 's/O representative/O to/O the/O European/**BO** Union/**IO** 's/O
veterinary/O committee/O Werner/**BP** Zwingman/**IP** said/O on/O Wednesday/O
consumers/O should/O...

Sequence Modeling in NLP

- ▶ Annotating a sequence of text tokens with a sequence of labels
 - ▶ Task 1: Part-of-speech Tagging
 - ▶ Task 2: Named entity Recognition
- ▶ Next step: Modeling a sequence of text tokens

Sequence Modeling

- ▶ Sequence Modeling in NLP
- ▶ **Language Modeling Task**
- ▶ Recurrent Neural Network
- ▶ Attention
- ▶ Transformer

Language Model Task: Examples

It is very hot during the summer in **Texas**.

vs

It is very hot during the summer in **London**.

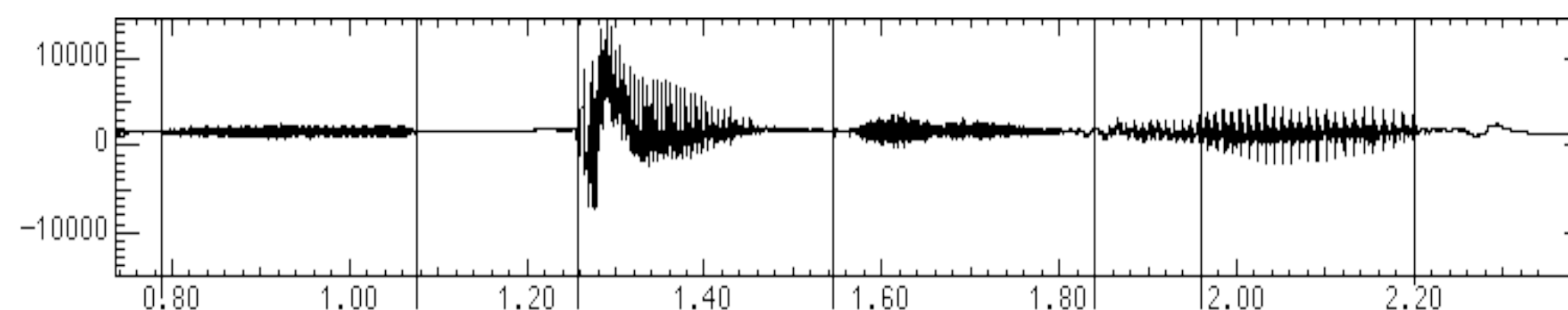
- ▶ Both are valid English sentences
- ▶ Which sentences are more *likely*?
- ▶ Goal: Assign probabilities $p(x)$ to sentences x

Why do we care?

- ▶ LLM-based chatbots, coding agents, etc.

Why do we care?

- ▶ Automatic speech recognition:
 - ▶ Speech in, Text out



Wreck a nice beach
vs.
Recognize speech

Eye eight uh Jerry
Vs.
I ate a cherry



Why do we care?

- ▶ Useful for a wide range of text generation tasks:
 - ▶ automatic speech recognition
 - ▶ translation
 - ▶ dialogue
 - ▶ text simplification
 - ▶ paraphrasing
 - ▶ grammatical error correction
- ▶ Providing a learning objective to get better representation for words

The Language Modeling Task

- ▶ Setup: Assume a (finite) vocabulary of words, (infinite) set of sentences.

$$\mathcal{V} = \{\text{the, a, man, telescope, Beckham, two, Madrid, ...}\}$$

$$\mathcal{V}^\dagger = \{\text{the, a, the a, the fan, the man, the man with the telescope, ...}\}$$

- ▶ Data: a training set of example sentences
- ▶ Task: estimate a probability distribution over sentences

$$\sum_{x \in \mathcal{V}^\dagger} p(x) = 1$$

$$\text{and } p(x) \geq 0 \text{ for all } x \in \mathcal{V}^\dagger$$

$$p(\text{the}) = 10^{-12}$$

$$p(\text{a}) = 10^{-13}$$

$$p(\text{the fan}) = 10^{-12}$$

$$p(\text{the fan saw Beckham}) = 2 \times 10^{-8}$$

$$p(\text{the fan saw saw}) = 10^{-15}$$

...

The noisy channel model

- ▶ We want to predict a sentence given acoustics:

$$x^* = \operatorname{argmax}_x P(x|a)$$

- ▶ Noisy channel approach:

$$x^* = \operatorname{argmax}_x P(x|a)$$



Acoustic model: Distributions
over acoustic waves given a
sentence

Language model: Distributions
over sequences of words
(sentences)

Building Language Model

- ▶ Goal: Assign probabilities $P(\mathbf{x})$ to sentences $\mathbf{x}$
 - ▶ Input: many observations of training sentences $\mathbf{x}$
 - ▶ Output: system capable of computing $P(\mathbf{x})$
- ▶ Simplest option: empirical distribution over the training sentences

$$p(\mathbf{x}) = \frac{\text{count}(\mathbf{x})}{N} \quad \mathbf{x} = x_1, x_2, x_3, \dots, x_{|\mathbf{x}|}$$

- ▶ Decompose the probability:

$$P(\mathbf{x}) = P(x_1)P(x_2 | x_1)P(x_3 | x_2, x_1) \dots$$

$$\begin{aligned} P(\text{the cat sat on the mat}) &= P(\text{the}) * P(\text{cat}|\text{the}) * P(\text{sat}|\text{the cat}) \\ &\quad * P(\text{on}|\text{the cat sat}) * P(\text{the}|\text{the cat sat on}) \\ &\quad * P(\text{mat}|\text{the cat sat on the}) \end{aligned}$$

Markov Assumption

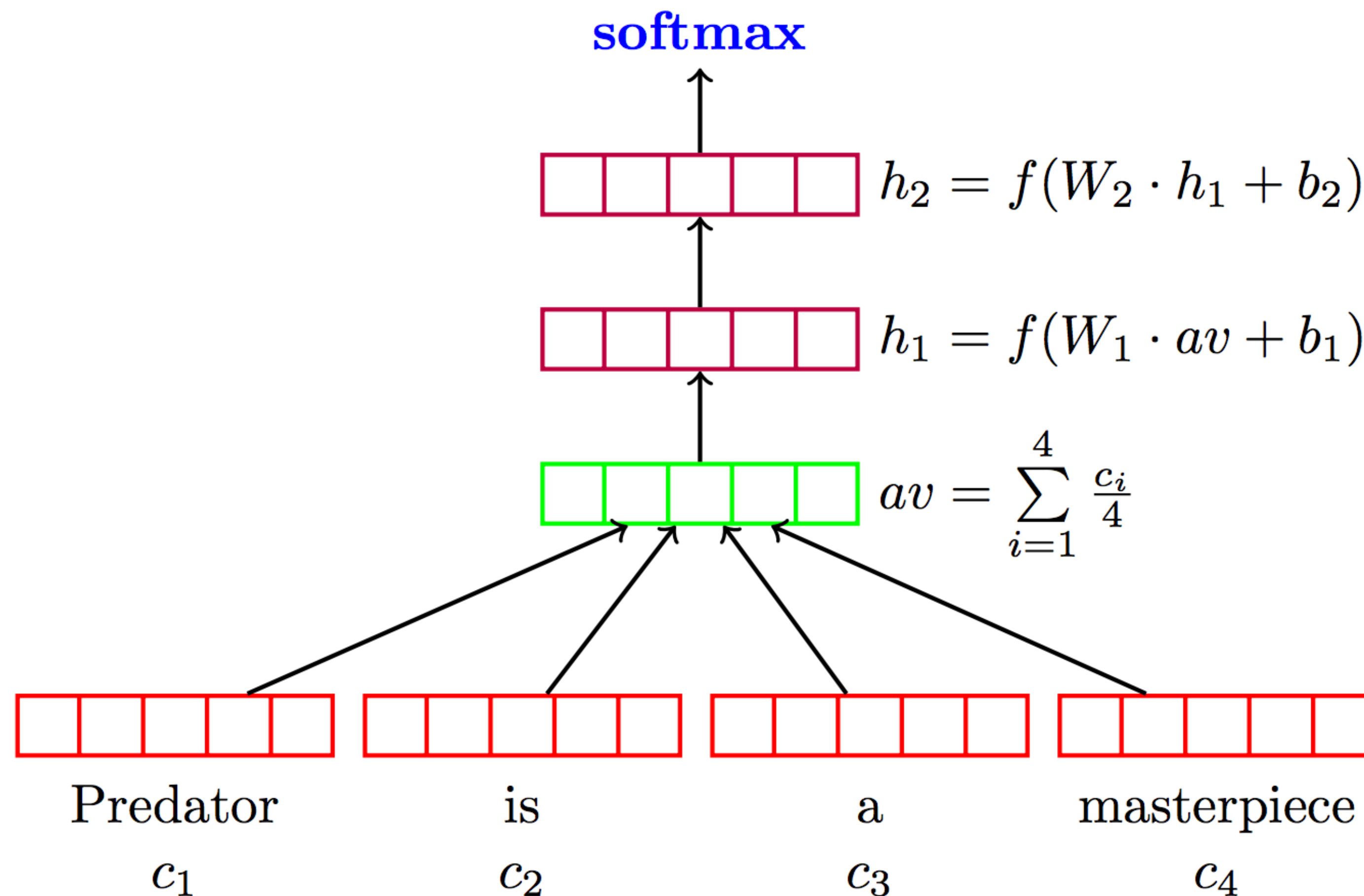
- ▶ Consider only the last k words as context

$$P(x_i | x_1, x_2, \dots, x_{i-1}) = P(x_i | x_{i-k} \dots x_{i-1})$$

$$P(x_1, x_2, \dots, x_{|\mathbf{x}|}) =$$

Recap: Deep Averaging Network

- Feedforward neural network on average of word embeddings from input

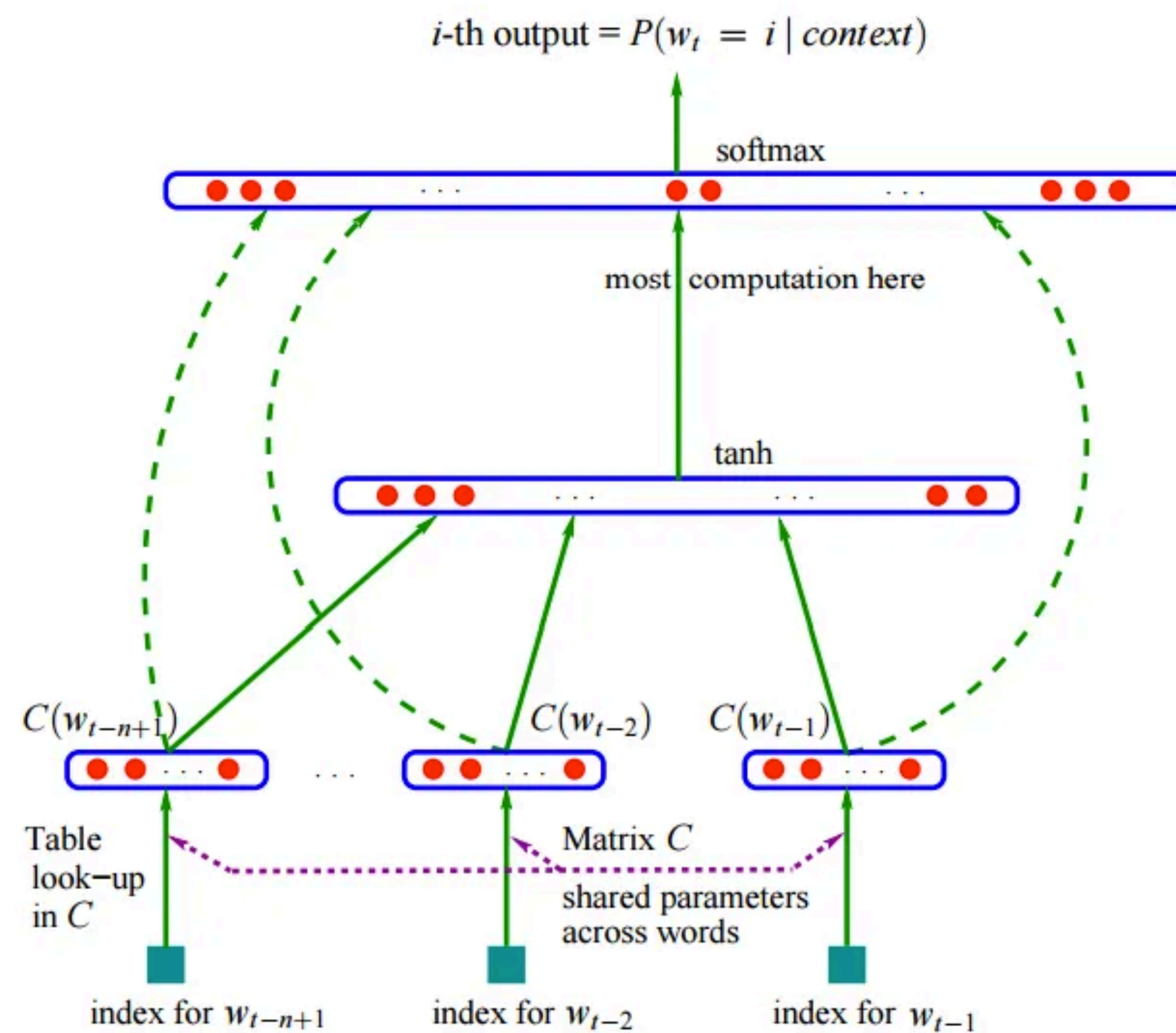


Iyyer et al. (2015)

Feedforward Neural Language Model



$$P(\text{mat} | \text{the cat sat on the})$$



- Input layer (context size $n = 5$):

$$\mathbf{x} = [\mathbf{e}_{\text{the}}; \mathbf{e}_{\text{cat}}; \mathbf{e}_{\text{sat}}; \mathbf{e}_{\text{on}}; \mathbf{e}_{\text{the}}] \in \mathbb{R}^{dn}$$

concatenation instead of averaging

- Hidden layer

$$\mathbf{h} = \tanh(\mathbf{W}\mathbf{x} + \mathbf{b}) \in \mathbb{R}^h$$

- Output layer (softmax)

$$\mathbf{z} = \mathbf{U}\mathbf{h} \in \mathbb{R}^{|V|}$$

$$P(w = i | \text{context}) = \text{softmax}_i(\mathbf{z})$$

Exercise:

1. What are the advantages of doing language modeling with a DAN?
2. What are the disadvantages?
3. What are the advantages of doing language modeling with a position-sensitive feedforward network?
4. What are the disadvantages?

Today

- ▶ Sequence Modeling in NLP
- ▶ Language Modeling Task
- ▶ **Recurrent Neural Network**
- ▶ Attention
- ▶ Transformer

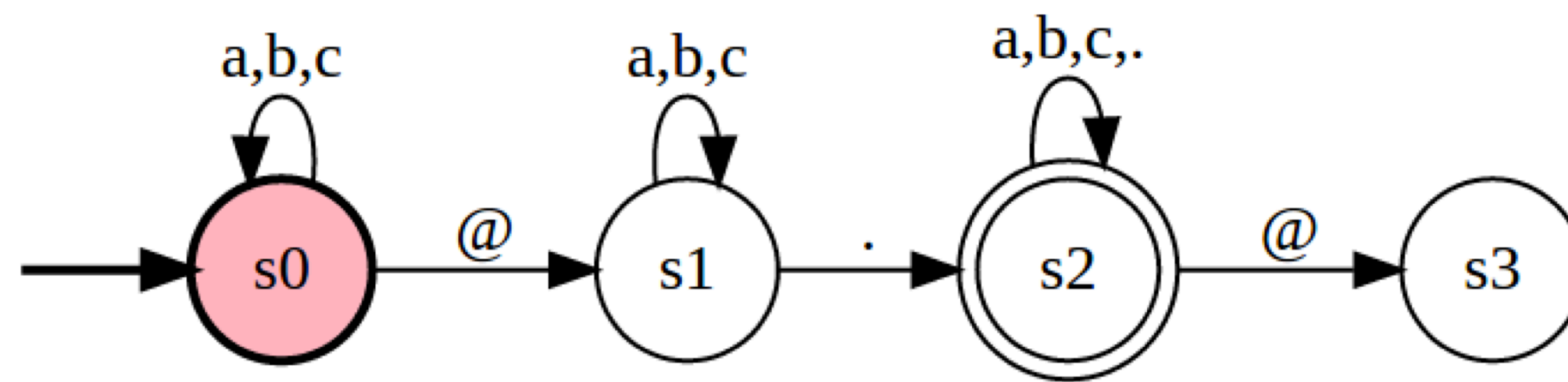
Why Sequence Modeling?

- ▶ Consider the following examples for sentiment classification:
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 - *You should not see this movie.*
- ▶ Would unigram, bigram models work?
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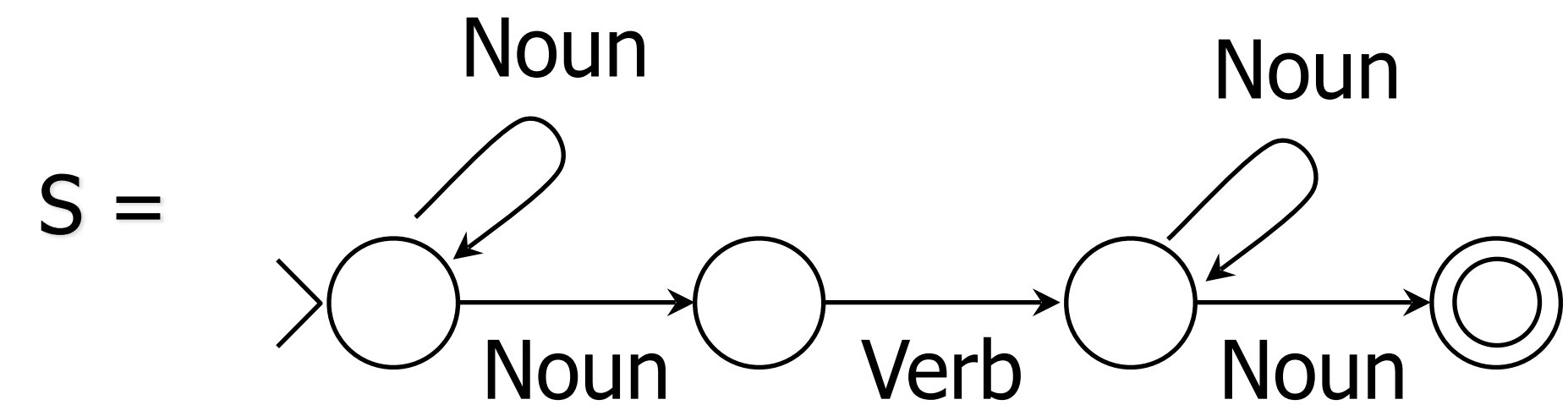
Classic Method: Finite State Machines

- ▶ Simple way of representing state in a sequential data
- ▶ Current state: saves necessary past information

- S – states
- Σ – vocabulary
- $s_0 \in S$ – start state
- $R: S \times \Sigma \rightarrow S$ – transition function



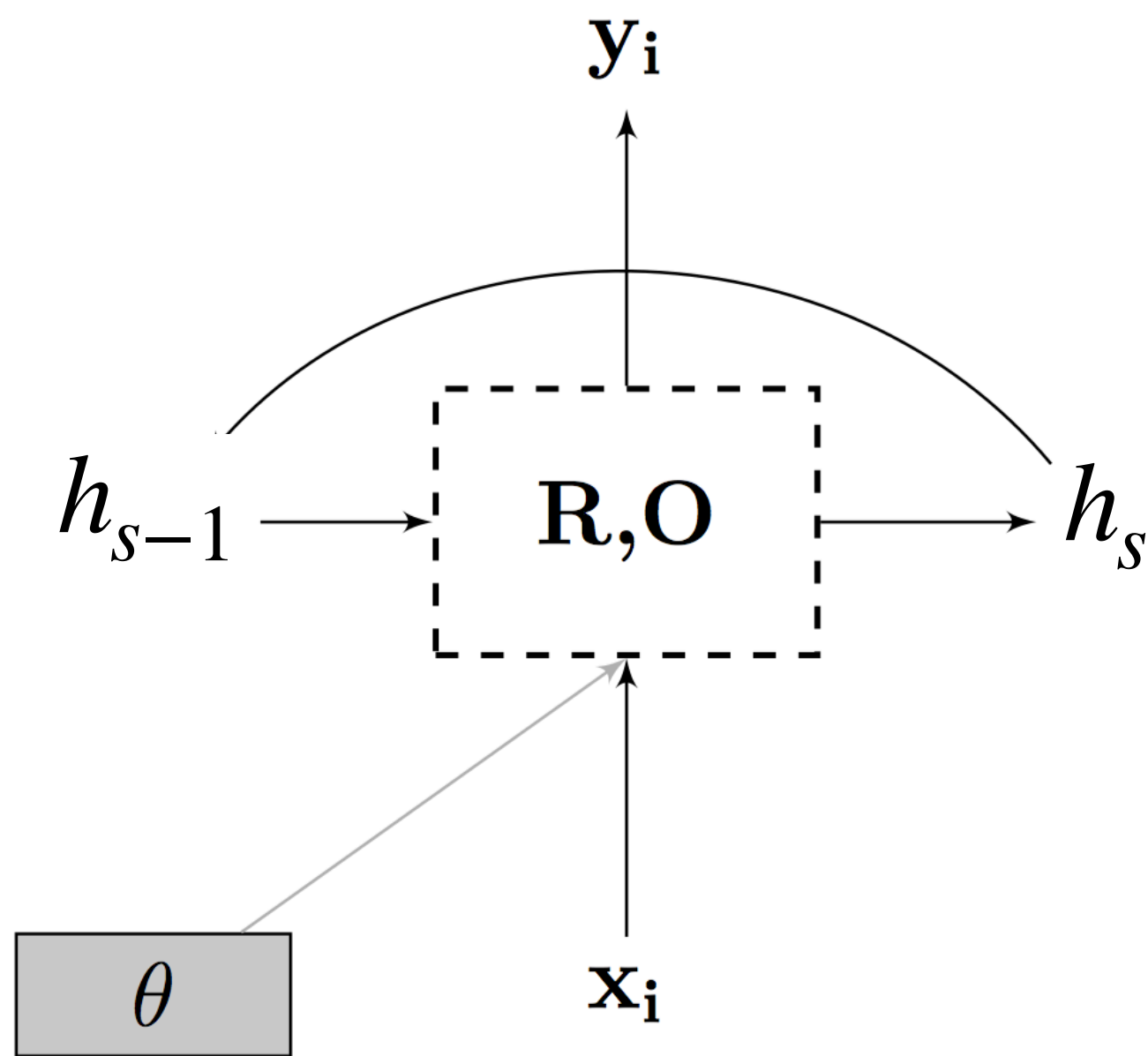
Email address parsing



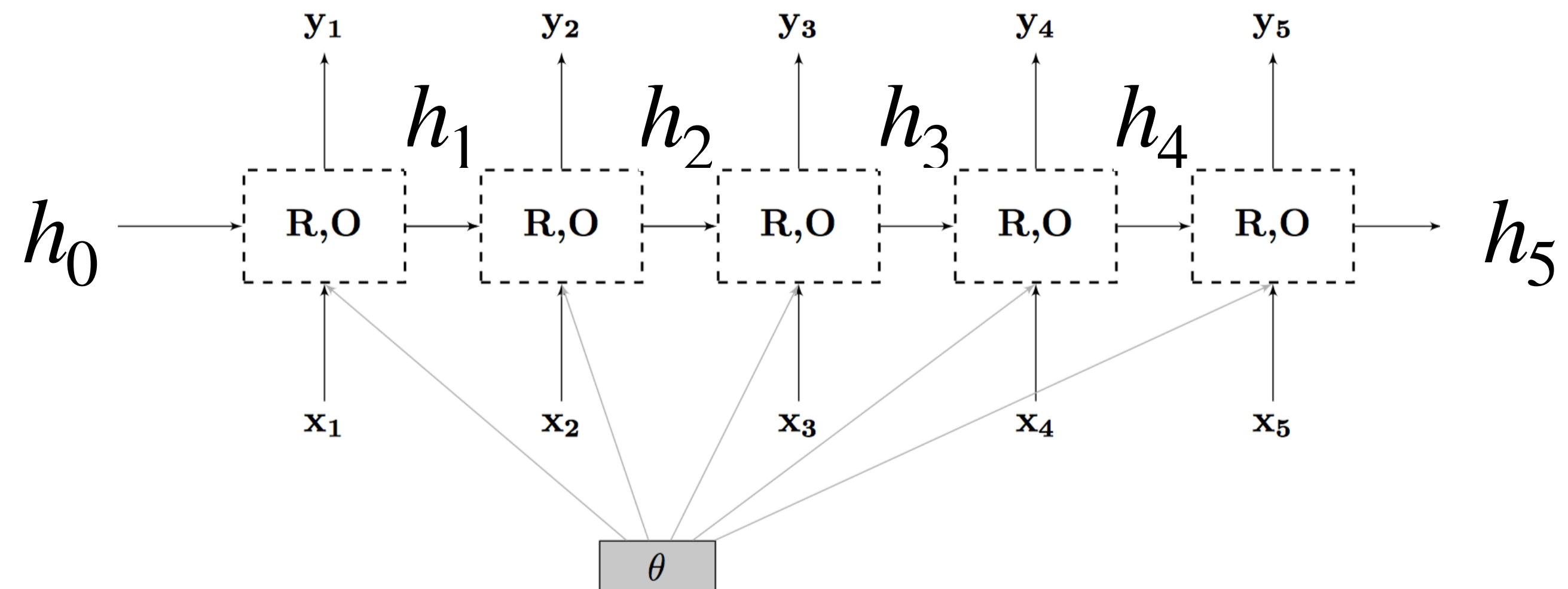
Language Model

Recurrent Neural Networks (RNN)

- ▶ Neural network model, but now with states!
- ▶ Handle variable length **inputs**



Recursive Representation



Unrolled Representation

RNNs

- ▶ Maps from dense input sequence to dense hidden state representation sequence

$$\mathbf{x}_1, \dots, \mathbf{x}_n \rightarrow h_1, \dots, h_n$$

- $S = \mathbb{R}^{d_{hid}}$ - hidden state space ($h_1, h_2 \dots$)
- $\Sigma = \mathbb{R}^{d_{in}}$ - input state space ($x_1, x_2 \dots$)
- $s_0 \in S$ - initial state vector (h_0)
- $R : \mathbb{R}^{d_{in}} \times \mathbb{R}^{d_{hid}} \rightarrow \mathbb{R}^{d_{hid}}$ - transition function

- ▶ For all $i \in \{1, \dots, n\}$,

- ▶ $h_i = R(h_{i-1}, \mathbf{x}_i)$

g : nonlinearity (e.g. tanh)

- ▶ Simple definition of R : $R(h_{i-1}, x_i) = g(Wh_{i-1} + Ux_i + b)$

- ▶ R is parameterized, where the parameters are shared across all steps.

$$h_4 = R(h_3, \mathbf{x}_4) = \dots = R(R(R(R(h_0, \mathbf{x}_1), \mathbf{x}_2), \mathbf{x}_3), \mathbf{x}_4)$$

RNNs

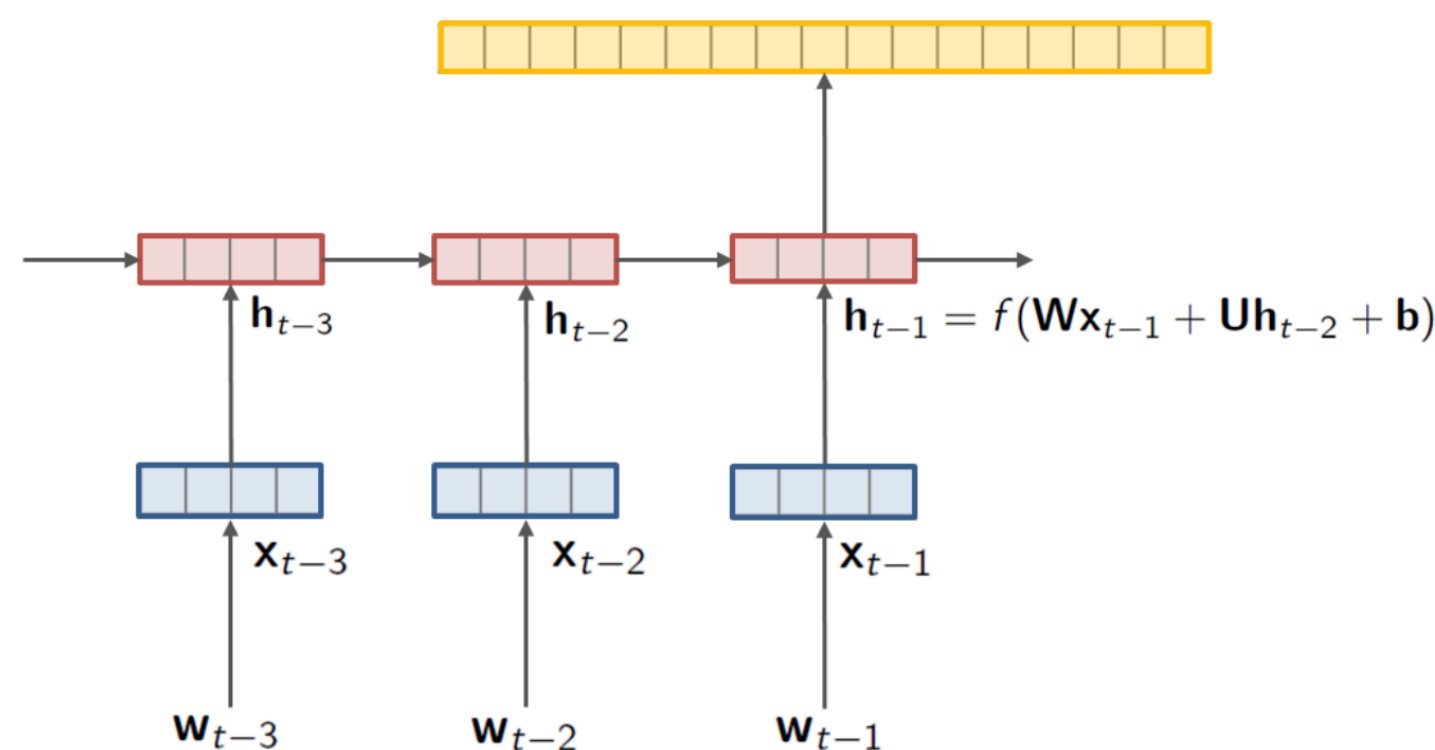
- ▶ Hidden states h_i can be used in different ways
- ▶ Output function maps hidden state vectors to symbols:

$$O: \mathbb{R}^{d_{hid}} \rightarrow \mathbb{R}^{d_{out}}$$

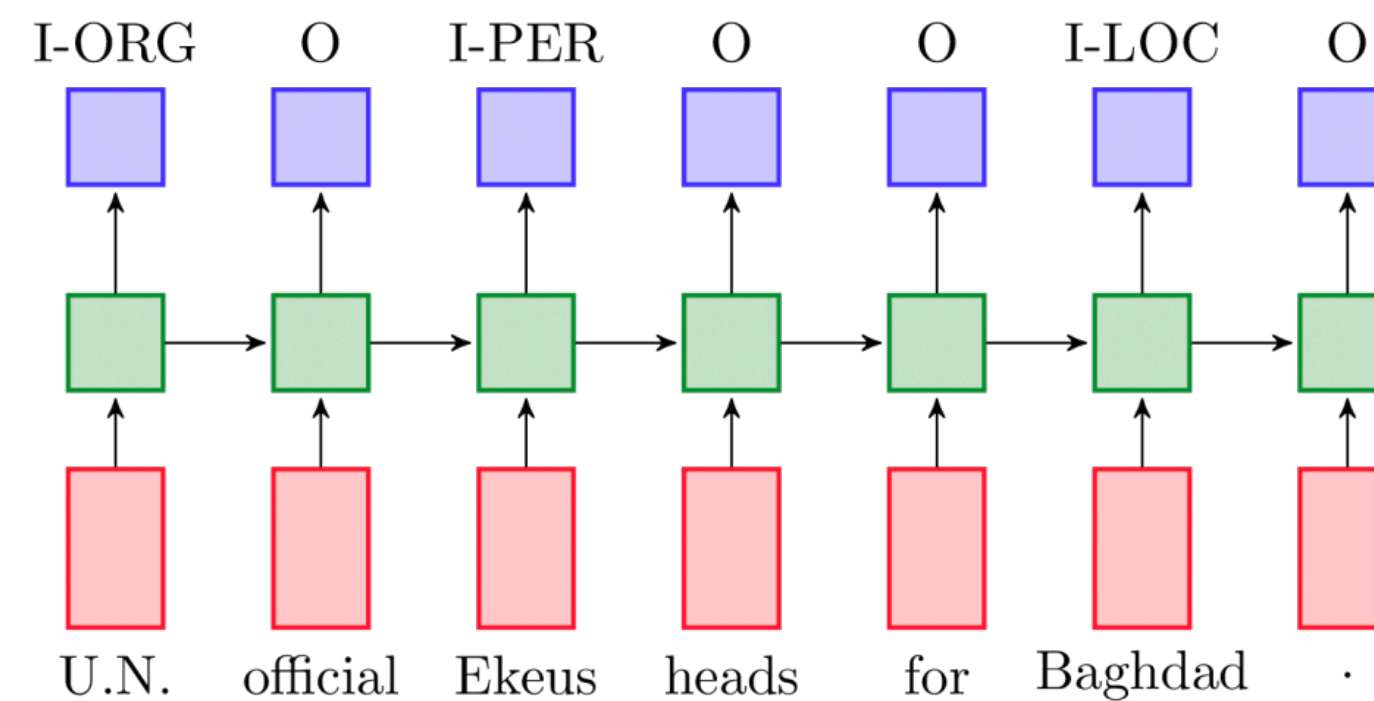
- ▶ For example: single layer + softmax

$$O(h_i) = \text{softmax}(h_i V + b_o)$$

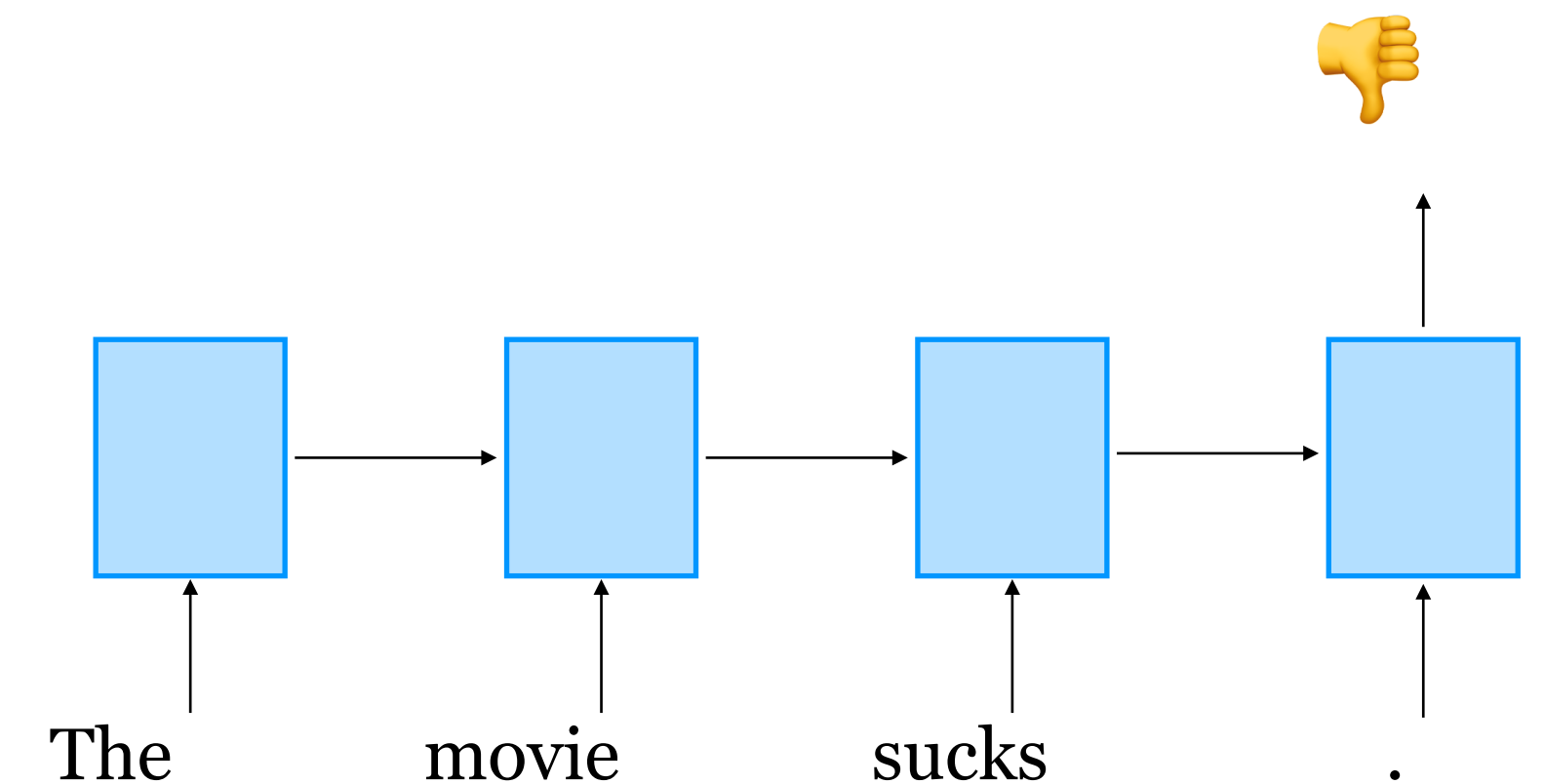
Language modeling



Sequence tagging

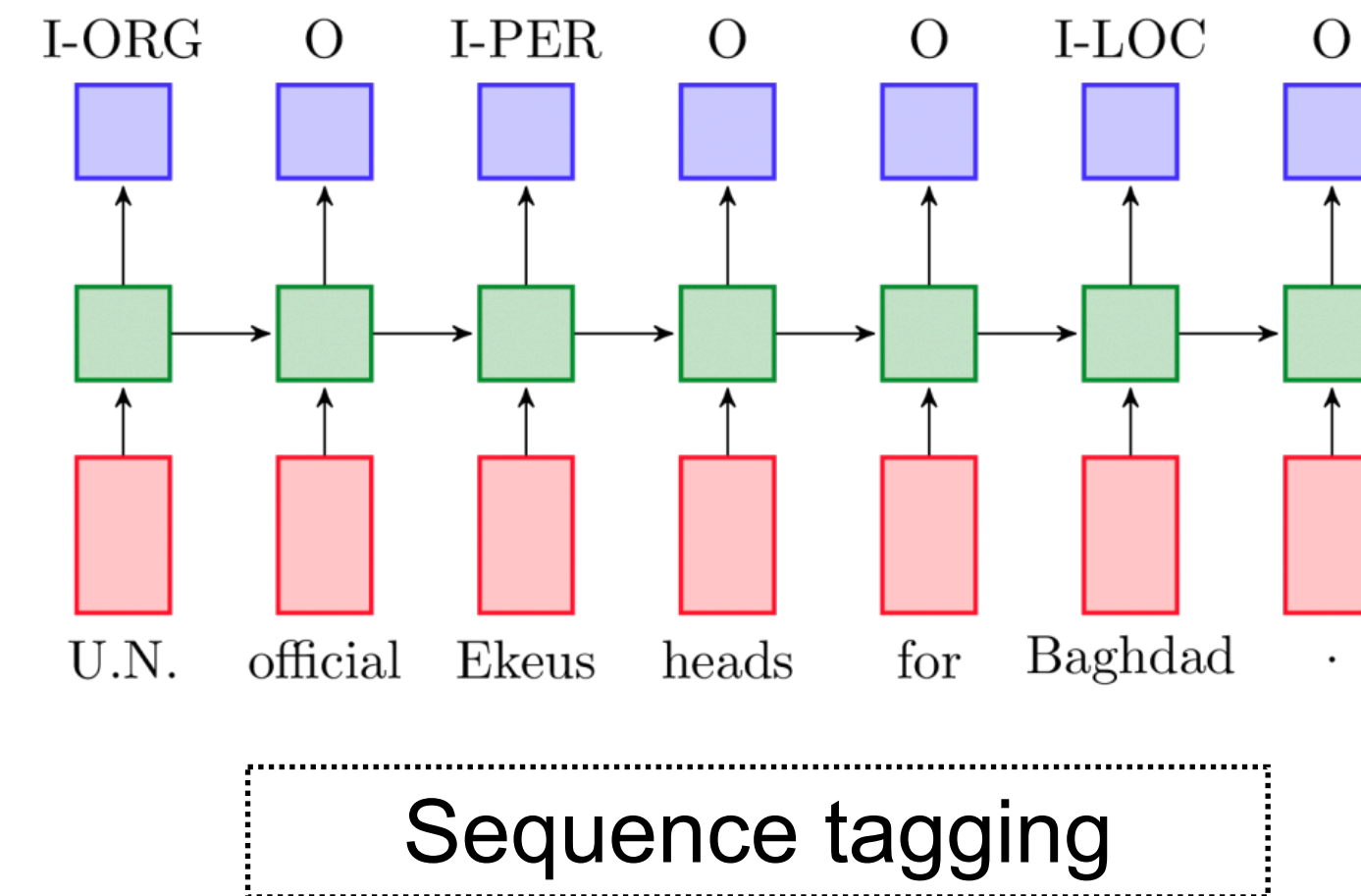
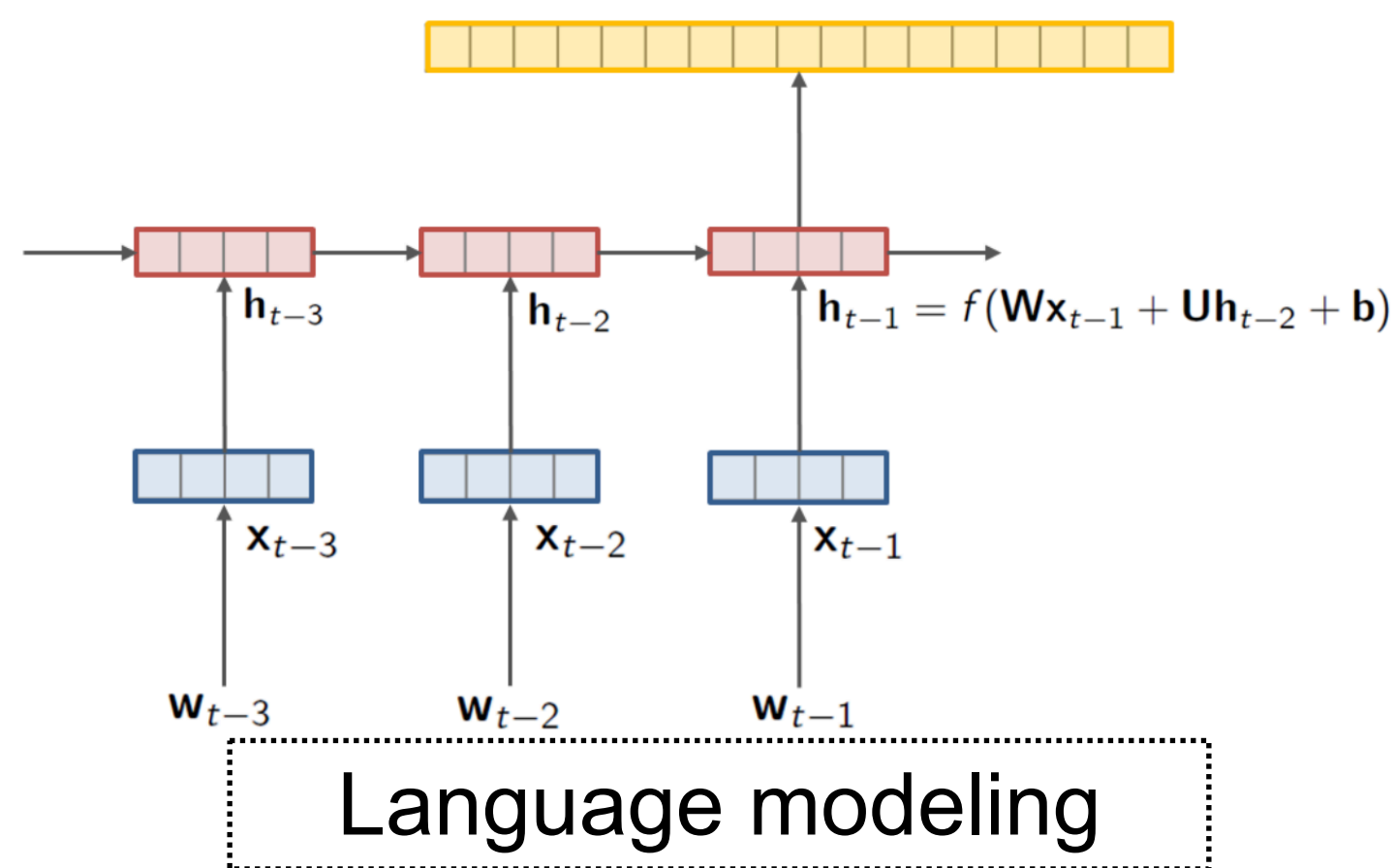


Text classification

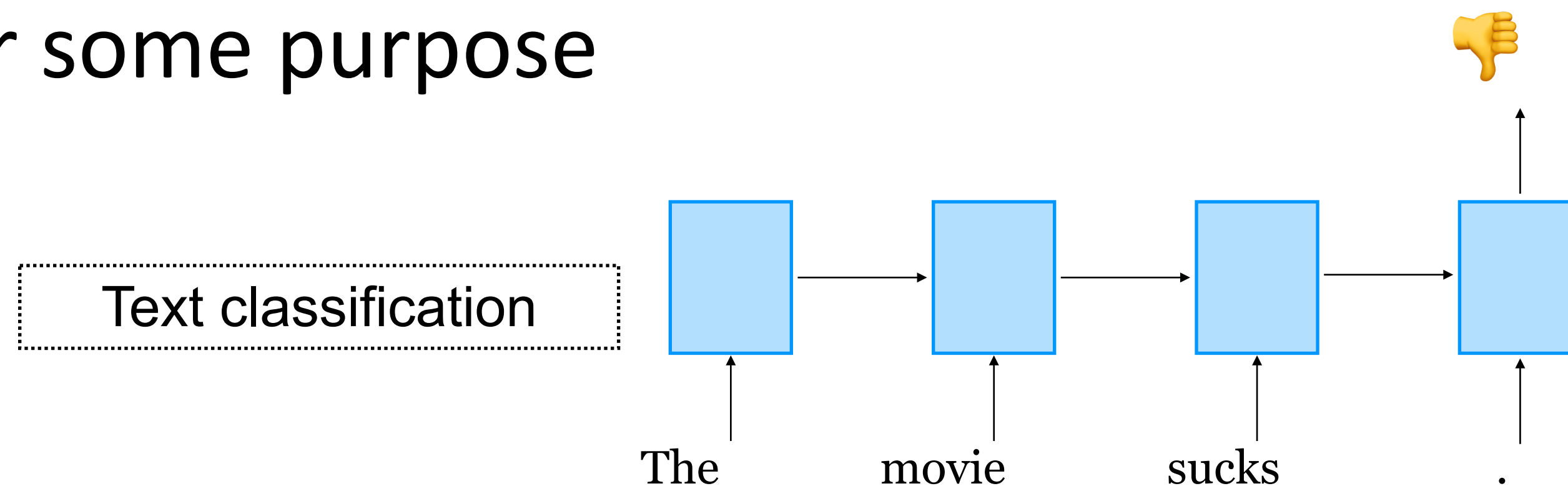


Output functions can be

- **Transducer**: make some prediction for each element in a sequence



- **Acceptor/encoder**: encode a sequence into a fixed-sized vector and use that for some purpose



Recurrent Neural Language Model

- Ingredients: a corpus of sequences of words $(x_1, x_2, x_3 \dots x_n)$

$$P(x_1, x_2, \dots, x_n) = P(x_1) \times P(x_2 | x_1) \times P(x_3 | x_1, x_2) \times \dots \times P(x_n | x_1, x_2, \dots, x_{n-1})$$
$$= P(x_1 | \mathbf{h}_0) \times P(x_2 | \mathbf{h}_1) \times P(x_3 | \mathbf{h}_2) \times \dots \times P(x_n | \mathbf{h}_{n-1})$$

- Loss function: Cross entropy-loss

- Predicted probability distribution, $p(x_t | h_{t-1})$, and true next word x_t
- You compute this loss over the entire training set

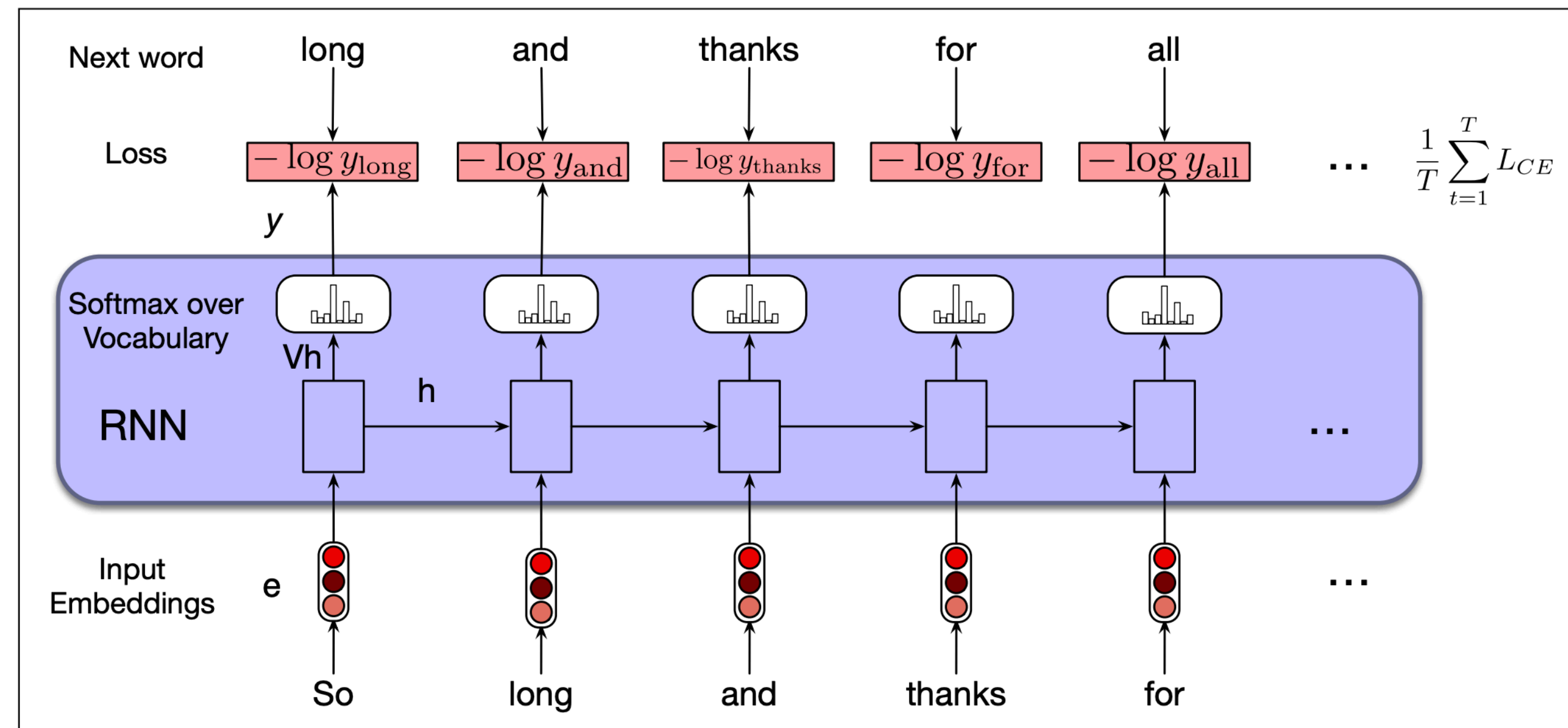
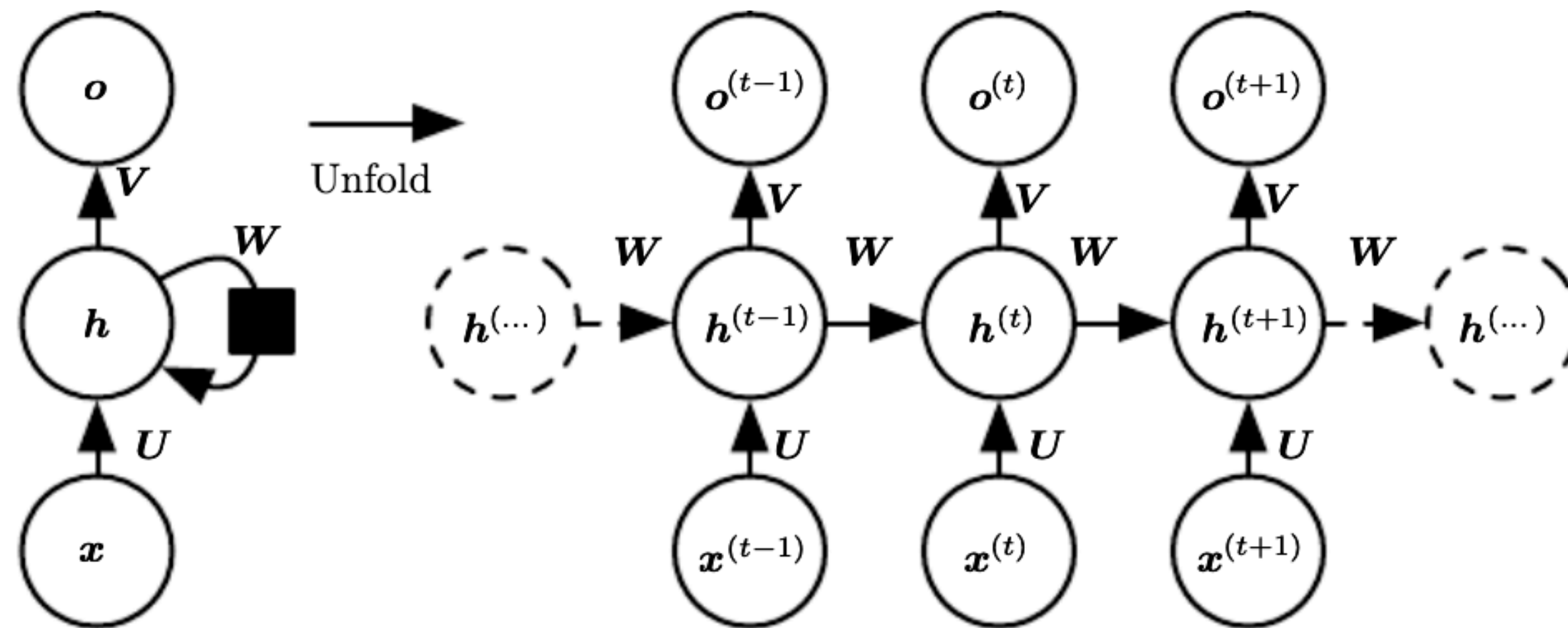


Figure 9.6 Training RNNs as language models.

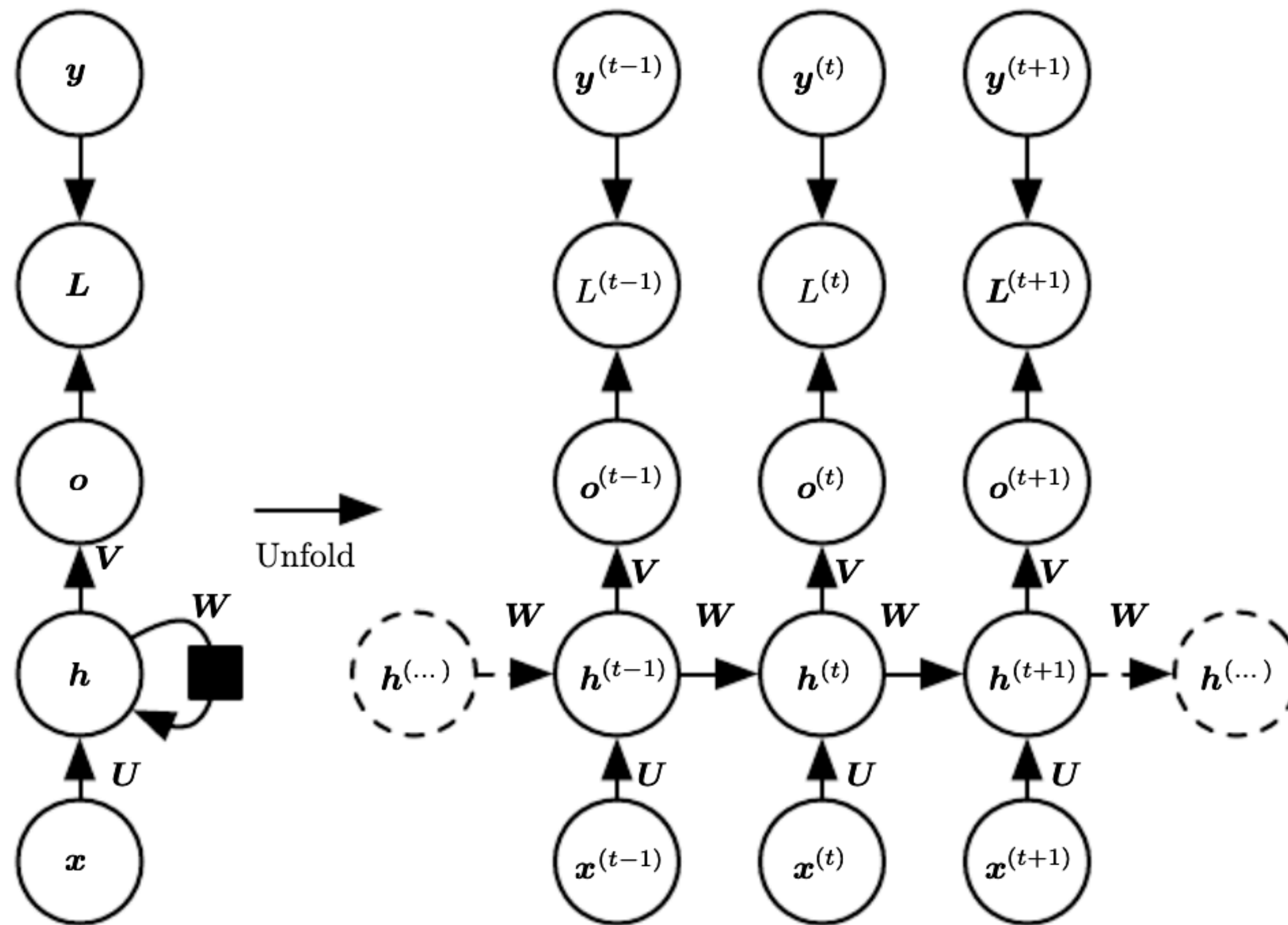
RNNs



$$R(h_{i-1}, x_i) = g(Wh_{i-1} + Ux_i + b)$$

$$O(h_i) = \text{softmax}(h_i V + b_o)$$

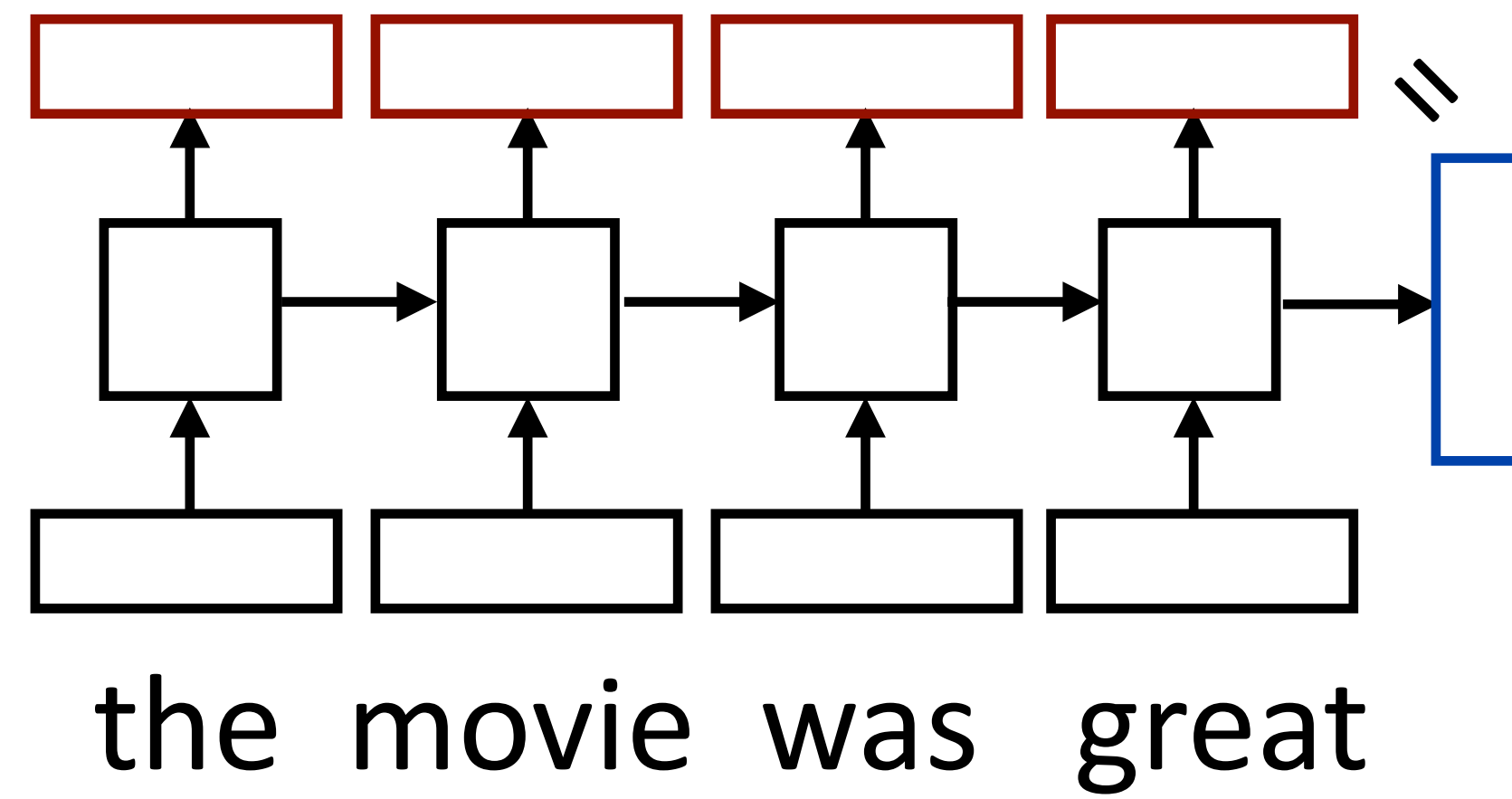
Training RNNs: Transducers



$$R(h_{i-1}, x_i) = g(Wh_{i-1} + Ux_i + b)$$

$$O(h_i) = \text{softmax}(h_i V + b_o)$$

What do RNNs produce?



- ▶ **Encoding of the sentence** — can pass this a decoder or make a classification decision about the sentence
- ▶ **Encoding of each word** — can pass this to another layer to make a prediction (can also pool these to get a different sentence encoding)
- ▶ RNN can be viewed as a transformation of a sequence of vectors into a sequence of context-dependent vectors

RNN can generate text!

Obama speeches

Good afternoon. God bless you.

The United States will step up to the cost of a new challenges of the American people that will share the fact that we created the problem. They were attacked and so that they have to say that all the task of the final days of war that I will not be able to get this done. The promise of the men and women who were still going to take out the fact that the American people have fought to make sure that they have to be able to protect our part. It was a chance to stand together to completely look for the commitment to borrow from the American people. And the fact is the men and women in uniform and the millions of our country with the law system that we should be a strong stretchs of the forces that we can afford to increase our spirit of the American people and the leadership of our country who are on the Internet of American lives.

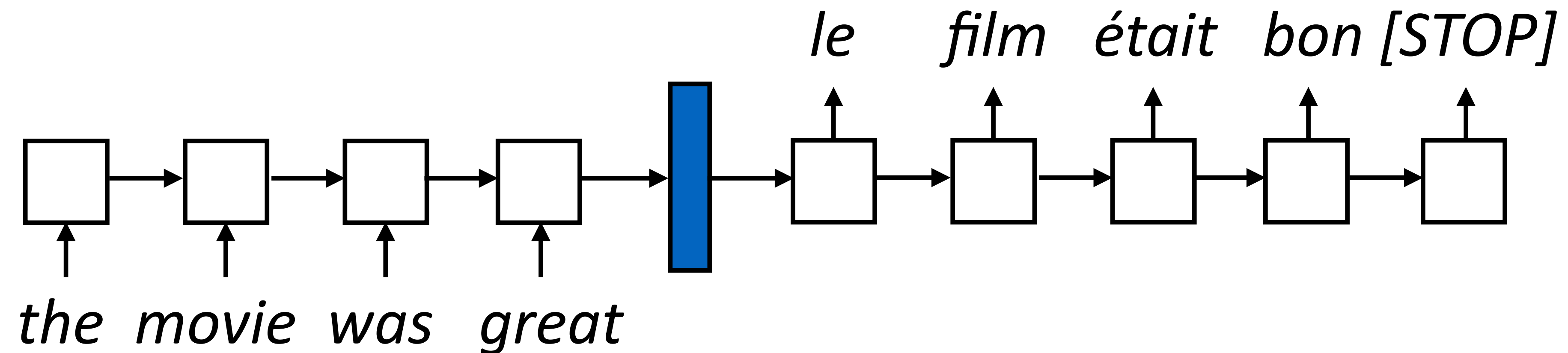
Thank you very much. God bless you, and God bless the United States of America.

Latex generation

```
\begin{proof}
We may assume that  $\mathcal{I}$  is an abelian sheaf on  $\mathcal{C}$ .
\item Given a morphism  $\Delta : \mathcal{F} \rightarrow \mathcal{I}$ 
is an injective and let  $q$  be an abelian sheaf on  $X$ .
Let  $\mathcal{F}$  be a fibered complex. Let  $\mathcal{F}$  be a category.
\begin{enumerate}
\item \hyperref[setain-construction-phantom]{Lemma}
\label{lemma-characterize-quasi-finite}
Let  $\mathcal{F}$  be an abelian quasi-coherent sheaf on  $\mathcal{C}$ .
Let  $\mathcal{F}$  be a coherent  $\mathcal{O}_X$ -module. Then
 $\mathcal{F}$  is an abelian catenary over  $\mathcal{C}$ .
\item The following are equivalent
\begin{enumerate}
\item  $\mathcal{F}$  is an  $\mathcal{O}_X$ -module.
\end{enumerate}
\end{enumerate}
\end{lemma}
```

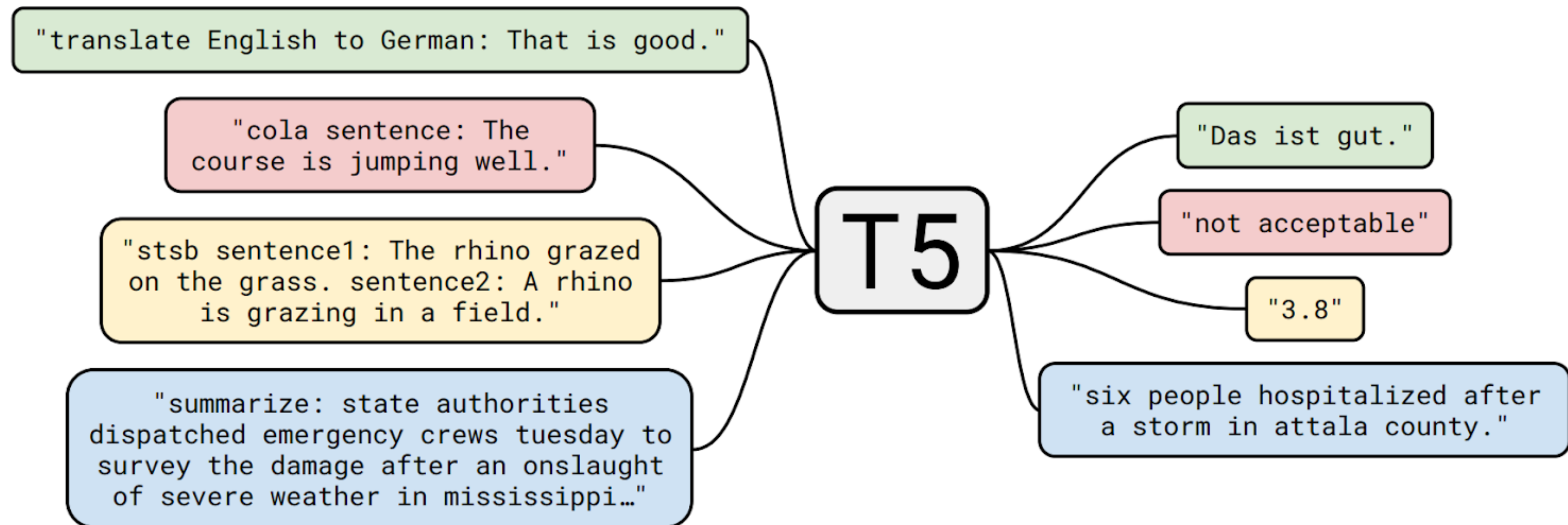
Seq2Seq Model w/ RNN

- ▶ Input: a sequence of tokens
- ▶ Output: a sequence of tokens (of *arbitrary* length)

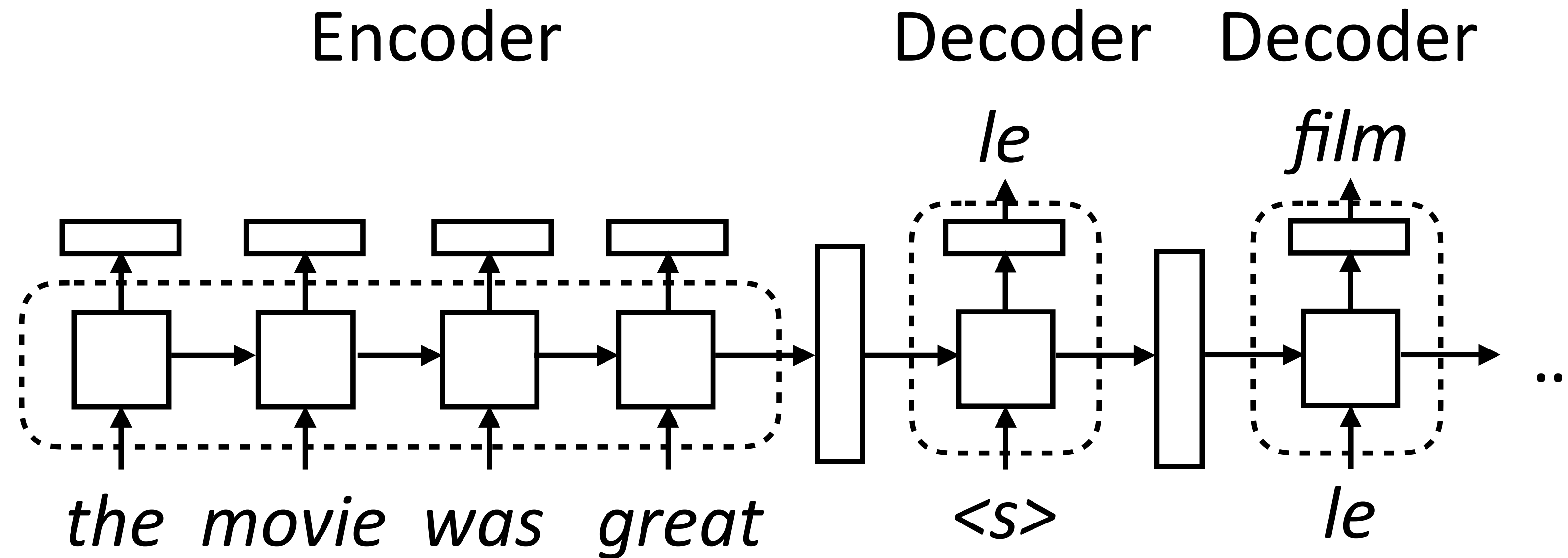


Text generation can embrace many tasks

- Conditional generation problem — given an input, generate an output as a text sequence

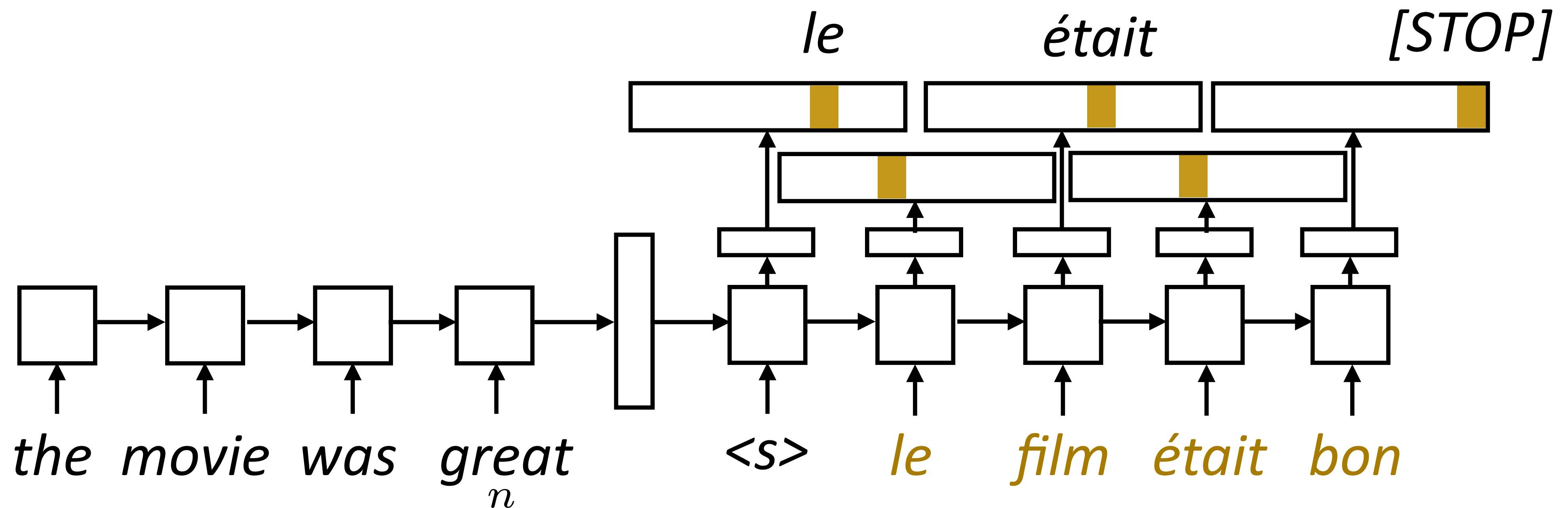


Seq2Seq Model w/ RNN



- ▶ Encoder: a RNN encoding a sequence of tokens, produces a vector.
- ▶ Decoder: separate RNN module (**different** parameters).
 - ▶ Takes two inputs: hidden state and previous token.
 - ▶ Outputs token and a new hidden state.

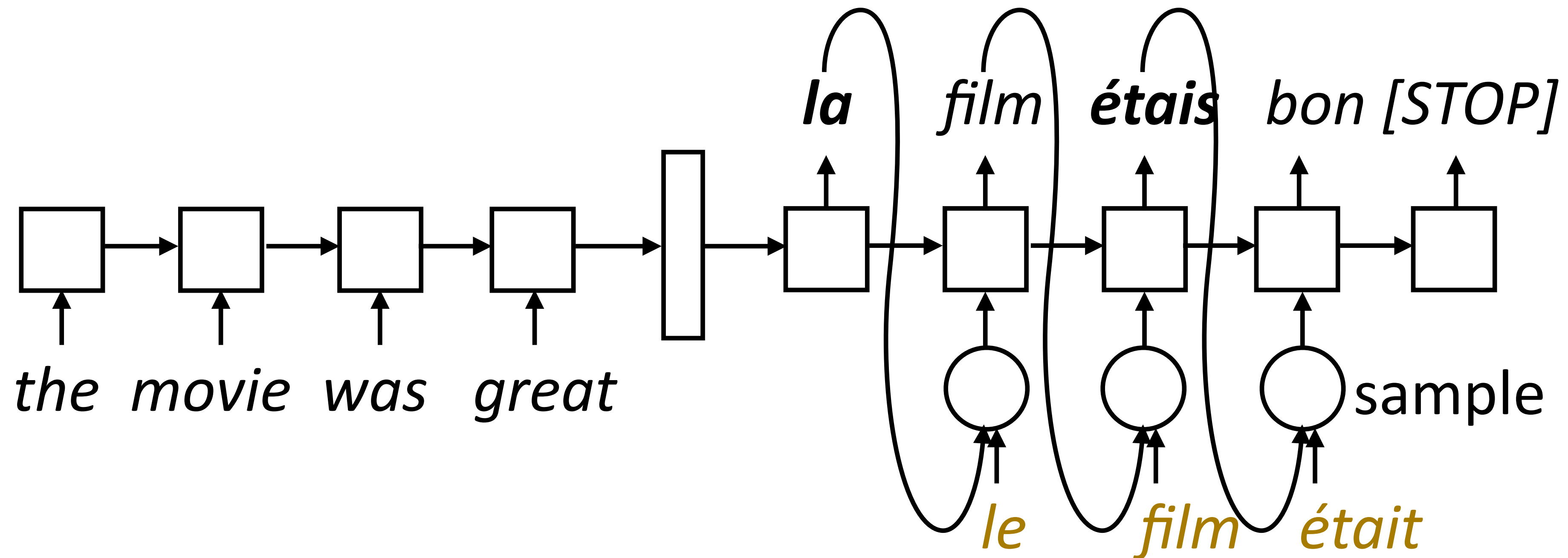
Training



- ▶ Objective: maximize $\sum_{(\mathbf{x}, \mathbf{y})} \sum_{i=1}^n \log P(y_i^* | \mathbf{x}, y_1^*, \dots, y_{i-1}^*)$
- ▶ One loss term for each target-sentence word, feed the correct word regardless of model's prediction (called "teacher forcing")
- ▶ Encoder and decoder parameters trained together! "End-to-End" training

Training: Scheduled Sampling

- Model needs to do the right thing even with its own predictions



- Scheduled sampling: with probability p , take the gold as input, else take the model's prediction
- Starting with $p = 1$ (teacher forcing) and decaying it works best